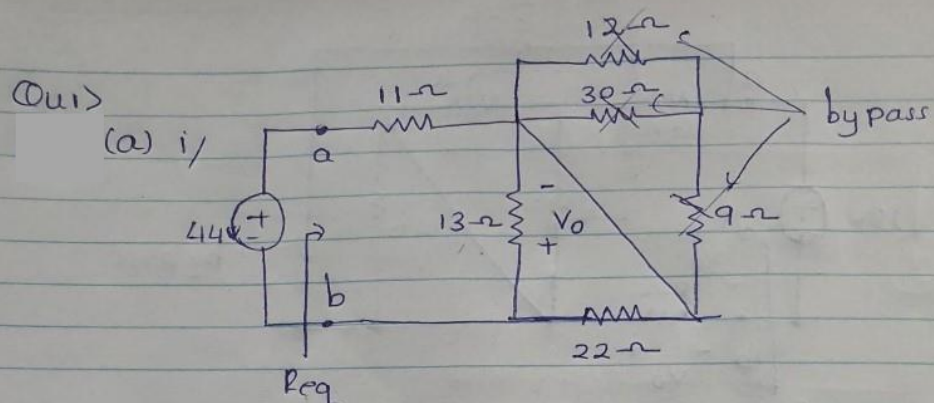
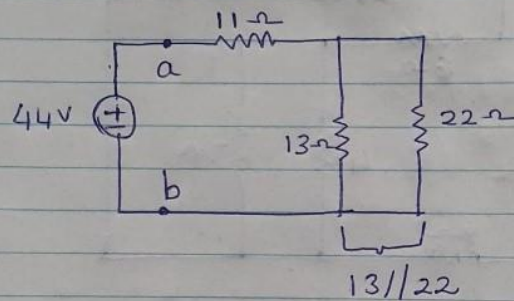


QUESTION 1a



if we re-draw the circuit



$$\begin{aligned} R_{eq} &= 11 + [13 // 22] \\ &= 11 + 8.17 \\ &= 19.17 \Omega \end{aligned}$$

ii/

using voltage division law

$$\begin{aligned} V_0 &= \frac{-8.17}{19.17} \times 44 \\ &= -18.75 \text{ V} \end{aligned}$$

QUESTION 1b

Q1(b)

Coffee maker = C.M
Micro wave = M.W
Toaster = T.O

120V

C.M.

M.W.

T.O.

$I_{C.M.}$

$I_{M.W.}$

$I_{T.O.}$

$I_{max} = 15A$

$I_{C.M.} = \frac{P}{V} = \frac{800}{120} = 6.67A$

$I_{M.W.} = \frac{1200}{120} = 10A$

$I_{T.O.} = \frac{850}{120} = 7.08A$

$I_{C.M.} + I_{M.W.} = 16.67A > 15A$

$I_{C.M.} + I_{T.O.} = 13.75A < 15A$

$I_{M.W.} + I_{T.O.} = 17.08A > 15A$

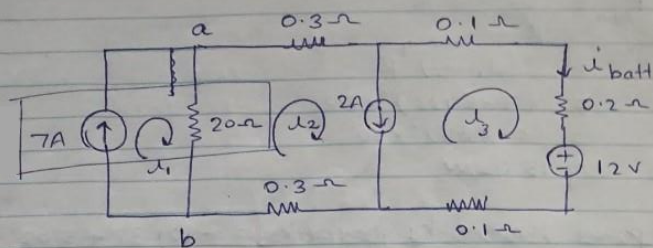
$I_{C.M.} + I_{M.W.} + I_{T.O.} = 23.75A > 15A$

$\therefore$ C.M. and T.O. operate simultaneously

QUESTION 2

Qu2

(a)



$$u_1 = 7A \rightarrow (1)$$

$$u_2 - u_3 = 2A \rightarrow (2)$$

$$0.6 u_2 + 20(u_2 - u_1) + 0.4 u_3 + 12 = 0 \rightarrow (3)$$

$$20.6 u_2 - 20 u_1 + 0.4 u_3 + 12 = 0$$

$$20.6(2 + u_3) - 140 + 0.4 u_3 + 12 = 0$$

$$21 u_3 = 140 - 12 - 41.2$$

$$21 u_3 = 86.8$$

$$u_3 = 4.133 A$$

$$u_{\text{batt}} = u_3 = 4.133 A //$$

(b) Total energy - battery =

$$W = V \times \Delta q$$

$$= 12 \times 4.133 \times (3600 \times 8)$$

$$= 1.42 \times 10^6 J //$$

(c)

$$P_{7A} = -V \cdot I$$

$$= [20 \times (u_1 - u_2)] \times 7$$

$$= -20 \times (7 - 6.133) \times 7$$

$$= -121.38 W$$

$$u_2 = 2 + u_3$$

$$= 2 + 4.133$$

$$= 6.133 //$$

$$P_{20\Omega} = \frac{V^2}{R} = I^2 R$$

$$= (u_1 - u_2)^2 R$$

$$= (7 - 6.133)^2 \times 20$$

$$= (0.867)^2 \times 20$$

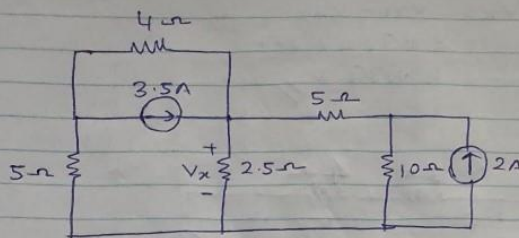
$$= 15.03 W$$

$$-121.38 + 15.03$$

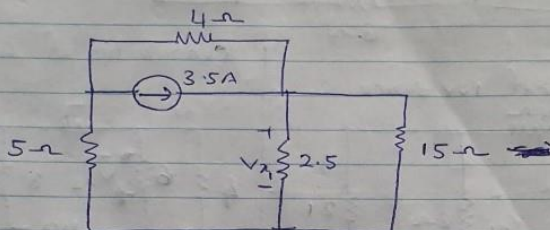
$$\text{Power - Solar} = -106.34 W //$$

QUESTION 3a

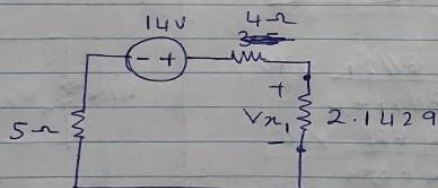
Qu3



Contribution of 3.5A current source



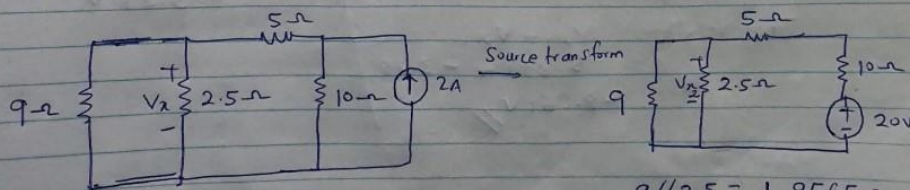
Source transformation



$$V_{x1} = \frac{2.1429}{5 + 4 + 2.1429} \times 14$$

$$= +2.692 \text{ V}$$

Contribution of 2A Current source



$$9 // 2.5 = 1.9565 \Omega$$

$$V_{x2} = \frac{1.9565}{1.9565 + 5 + 10} \times 20$$

$$= 2.307 \text{ V}$$

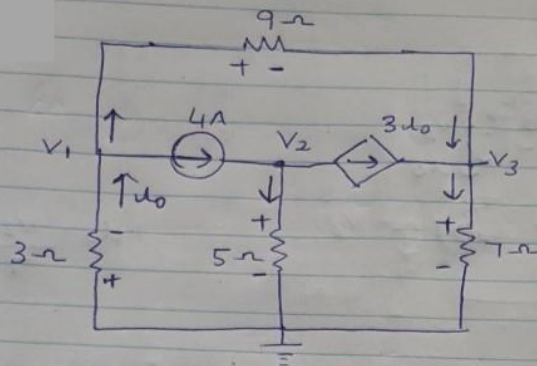
Final results $V_x = V_{x1} + V_{x2}$

$$= 2.692 + 2.307$$

$$= 4.999 \text{ V} //$$

QUESTION 3b

Q43b



KCL @ node 1

$$u_o - \frac{(V_1 - V_3)}{9} - 4 = 0 \quad ; \quad u_o = \frac{0 - V_1}{3}$$

$$-\frac{V_1}{3} - \frac{V_1}{9} + \frac{V_3}{9} = 4$$

$$\frac{V_3}{9} - \frac{4V_1}{9} = 4 \quad \text{--- (1)}$$

KCL @ node 2

$$4 - \frac{V_2}{5} - 3u_o = 0$$

$$4 - \frac{V_2}{5} + \frac{3V_1}{3} = 0$$

$$\frac{V_2}{5} - V_1 = 4 \quad \text{--- (2)}$$

KCL @ Node 3

$$\frac{V_1 - V_3}{9} + 3u_o - \frac{V_3}{7} = 0$$

$$\frac{V_1}{9} - \frac{V_3}{9} - \frac{V_3}{7} = -V_1 = 0$$

$$\frac{8V_1}{9} + \frac{16}{63}V_3 = 0 \quad \text{--- (3)}$$

Nodal eq.:

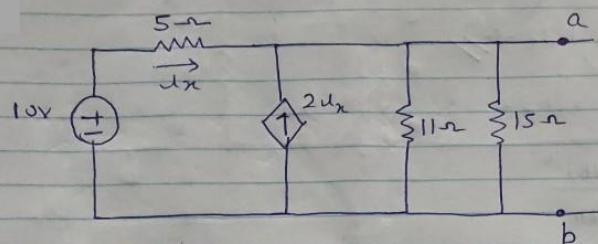
$$\frac{-4V_1}{9} + \frac{V_3}{9} = 4$$

$$-V_1 + \frac{V_2}{5} = 4$$

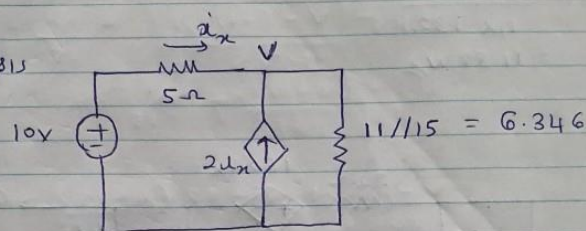
$$\frac{8V_1}{9} + \frac{16}{63}V_3 = 0$$

QUESTION 4

Q44)



Nodal analysis



$$u_x + 2u_x - \frac{V}{6.346} = 0 \quad \text{where } u_x = \frac{10 - V}{5}$$

$$= 2 - \frac{V}{5}$$

$$\frac{2 - V}{5} + 4 - \frac{2V}{5} - \frac{V}{6.346} = 0$$

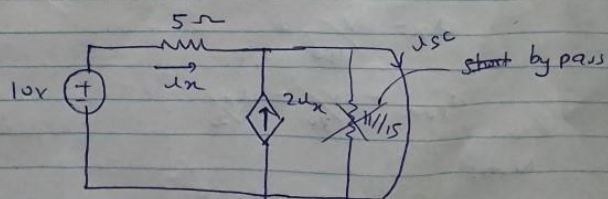
$$6 - \frac{3V}{5} - \frac{V}{6.346} = 0$$

$$\frac{(3 \times 6.346 + 5)V}{5 \times 6.346} = 6$$

$$V = 7.92V$$

$$V_{th} = 7.92V //$$

$$R_{th} = ? = \frac{V_{th}}{I_{sc}}$$



$$I_{sc} = u_x + 2u_x$$

$$= 3u_x$$

$$= 6A$$

$$u_x = \frac{(10 - 0)}{5} = 2A$$

$$R_{th} = \frac{7.92}{6}$$

$$= 1.32\Omega //$$

(b) $R_{light} = R_{th} = 1.32$

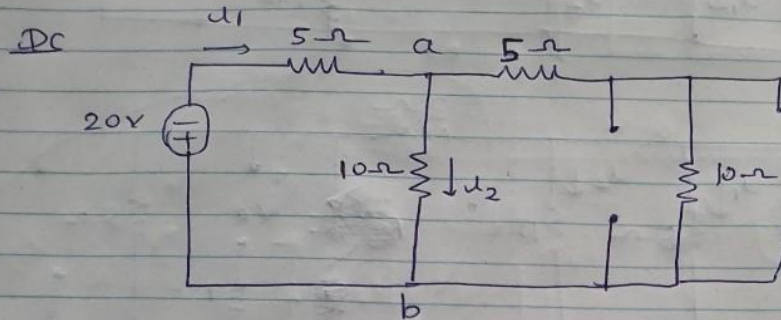
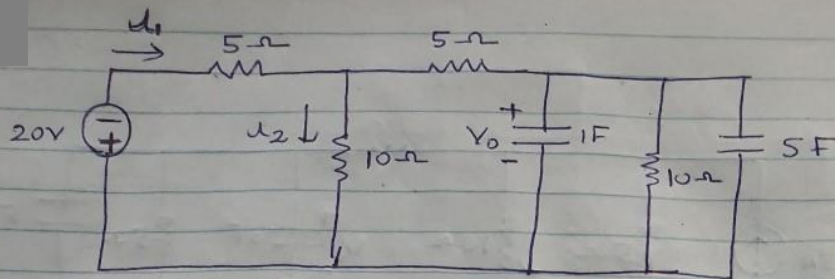
(c) $P_{max} = \frac{(V_{th})^2}{4R_{th}}$

$$= \frac{(7.92)^2}{4 \times 1.32}$$

$$= 11.88W //$$

QUESTION 5a

(Q5a)



$$I_1 = \frac{-20}{5 + 10 // [10 + 5]}$$

$$I_1 = \frac{-20}{5 + 10 // 15} = \frac{-20}{11} = -1.818 \text{ A} //$$

$$I_2 = \frac{R_{eq}}{10} \times I_1 \quad R_{eq} = 10 // 15 = 6$$

$$= \frac{6}{10} \times (-1.818)$$

$$= -1.090 \text{ A} //$$

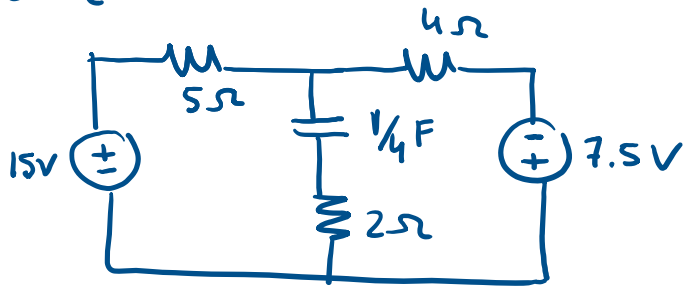
$$V_0 = \frac{10}{15} V_{ab} \quad ; \quad V_{ab} = -\frac{6}{6+5} \cdot 20 = -10.91$$

$$= \frac{10}{15} \times (-10.91) =$$

$$= -7.27 \text{ V} //$$

QUESTION 5b

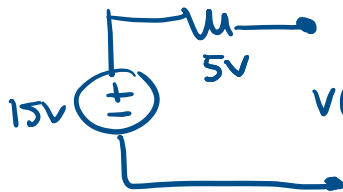
i) For $t > 0$



Step response

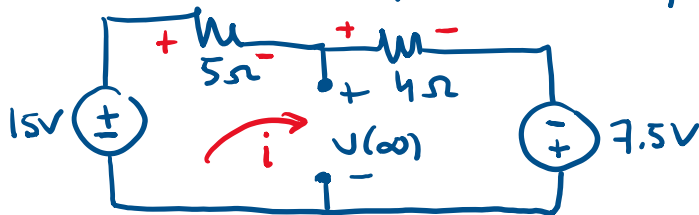
$$V(t) = V(\infty) + [V(0) - V(\infty)] e^{-t/\tau}$$
 where $\tau = R_{th} C$

* Calculate $V(0)$ for $t < 0$ (capacitor is o.c. in steady state)



$V(0) = 15V$

* Calculate $V(\infty)$ for $t > 0$ (capacitor is again o.c. in steady state)



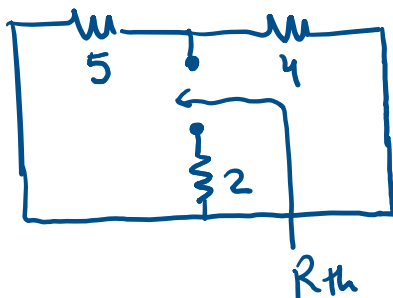
KVL @ full mesh: $-15 + 9i - 7.5 = 0$

$i = \frac{15 + 7.5}{9} = 2.5A$

KVL @ left mesh: $-15 + 5i + V(\infty) = 0$

$V(\infty) = 15 - 5i = 15 - 5 \times 2.5 = 2.5V$

* Calculate τ for $t > 0$



$R_{th} = 5 \parallel 4 + 2 = 4.22\Omega$

$\tau = R_{th} \times C = 4.22 \times 1/4 = 1.055$

$$V(t) = 2.5 + [15 - 2.5] e^{-t/1.055} = 2.5 + 12.5 e^{-t/1.055} \quad \checkmark$$

ii) $V(0.2) = 2.5 + 12.5 e^{-0.2/1.055} = 12.84V$